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ORIGINAL PAPER

# Community assembly of Diptera following restoration of mined boreal bogs: taxonomic and functional diversity

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**Abstract** Peat mining causes major degradation to bogs and natural regeneration of these sites is slow and often incomplete. Thus, restoration is an important tool for re-establishing natural ecosystem properties (although perhaps not the original species pool) in mined bogs. Because faunal recovery cannot be taken for granted following plant restoration, we assessed community assembly of higher flies (Diptera: Brachycera) in previously mined bogs 7 years after restoration. Species assemblages in restored sites were compared to those in nearby natural and abandoned mined sites. The three treatment types did not differ significantly in overall species composition, suggesting high resilience to disturbance. However, species richness and evenness were generally lower in abandoned sites than restored and natural sites, which had similar abundance distributions, indicating that restoration enhanced recovery of species diversity and community structure. Functional traits (trophic group, body size) provided a different insight into the status of restored sites. Trophic and small size-class (<5 mm) composition in restored sites were similar to those in abandoned sites. However, high species richness estimates indicated that predators and saprophages successfully colonized restored sites. Species assemblages were mostly affected by coverage of bare peat, *Sphagnum* mosses and ericaceous shrubs; trophic assemblages were affected by variables directly linked to feeding habits. Our results suggest that active restoration is needed for the renewal of high species and trophic diversity, although it is clear from environmental conditions and functional traits that the restored sites are not yet fully functioning peatlands 7 years after restoration.

**Keywords** Peatlands · Colonization · Trophic assemblages · Biodiversity · Environmental conditions

## Introduction

Canada has approximately 170 million ha of peatlands, of which bogs are the dominant type in the south (Gorham 1990). These southern bogs are under anthropogenic pressure through horticultural peat extraction, particularly in the St. Lawrence Lowlands (Pellerin 2003). Peat mining leads to major ecosystem changes: the water table is lowered to dry the peat and to allow the surface to support vacuum extractors, vegetation is removed, and a fine layer of peat is aspirated each year (Gorham and Rochefort 2003). The substrate is degraded in sites abandoned after years of extraction (Schouwenaars 1993; Campbell et al. 2002; Holden et al. 2004).

Regeneration of abandoned mined bogs and re-establishment of community structure and function to pre-disturbance levels in restored sites may be slow and unpredictable, especially if colonist sources are distant or isolated. Thus, active restoration is an important tool for the re-establishment of ecosystem properties and functions (Rochefort et al. 2003; Prach and Hobbs 2008). The restoration approach used in Canada was developed to aid in

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the recovery of abiotic and biotic characteristics, re-establish a stable water table and a plant cover dominated by native peatland species. Drainage ditches are blocked to raise and stabilize the water table, fields are re-profiled to reduce runoff and fertilizer is applied. Plants are re-introduced by spreading fragments of *Sphagnum* and other mosses, cotton grass (*Eriophorum*, Cyperaceae) and ericaceous shrubs harvested from natural remnants or other bogs. The area is mulched with straw to reduce desiccation (Quinty and Rochefort 2003).

Because plants are actively re-introduced, most emphasis has been on monitoring their establishment in restored peatlands (Pellerin and Lavoie 1999; Campeau et al. 2004; Chirino et al. 2006). Although animals are not actively introduced in these restoration projects, colonization of birds (Desrochers et al. 1998), amphibians (Mazerolle 2005) and aquatic arthropods (Mazerolle et al. 2006) has been monitored. Microbial establishment has also been assessed (Andersen et al. 2006). There is little knowledge of how terrestrial arthropods react to restoration of peatland-associated plant communities and ecosystem properties, even though they are the most species-rich group of animals and have a significant impact on ecosystem processes. This study focused on higher flies (Diptera: Brachycera), a diverse and ecologically important group in peatlands. Higher flies exhibit great functional diversity, include a number of bog specialists, are species rich and abundant, and thus may display a broad array of responses to environmental changes (Blades and Marshall 1994; Keiper et al. 2002; Spitzer and Danks 2006).

Analyses should consider functional groups, not just taxonomic groups, because such comparisons can reveal different effects of human activities and environmental conditions (Petchey and Gaston 2006). Changes in plant quality and quantity are likely to influence phytophagous insect diversity and this should cascade up to higher trophic levels, by indirectly or directly affecting the diversity of parasites and predators (Hunter and Price 1992; Siemann et al. 1998; Brose 2003). Saprophagous species may be influenced by the rate of decomposition and soil characteristics of the habitat (Rotheray et al. 2001; Keiper et al. 2002).

Because establishment of species in restored sites depends on biotic and abiotic habitat constraints, the main objectives of this study were to determine the effects of peatland restoration on species diversity (incorporating both species richness and relative abundance), functional diversity and species composition (relative abundance of different species) of Brachycera in natural, abandoned-mined and restored bogs in southeastern Canada; and the extent to which Brachycera diversity is associated with abiotic and biotic environmental variables within the treatments.

## Materials and methods

### Study area and sampling sites

The study was conducted in three lowland bogs in the St. Lawrence River estuary, Quebec that have been, or are still being, mined using the vacuum technique (Rochefort 2001). Each site contains a section that was abandoned after peat extraction, a section restored in 1999–2000 by the Peatland Ecology Research Group and a natural section without obvious disturbance. Bois-des-Bel bog (47°57′N, 69°25′W) covers 187 ha. A 11.5 ha section was mined from 1972 to 1980. One 8 ha section consisting of eight 30 × 240 m peat fields was restored in 1999–2000, and an abandoned area of 3 ha (two peat fields of 30 × 240 m) was kept as a control zone. Chemin-du-Lac bog (47°45′N, 69°31′W) is part of the Rivière-du-Loup peatland, the largest bog in this region. Several sectors are still mined. A 12.8 ha area abandoned in 2000 was used in this study. Eight sections were restored from 1997 to 2004; those restored in 1999 (3 ha) and 2000 (4 ha) were used in this study. St-Charles bog (46°45′N, 70°59′W) has an area of 1,306 ha with sections still mined. A section abandoned in 1986 was used in this study. One 30 × 220 m peat field was restored in 1999.

### Insect sampling and processing

Bogs were sampled from 01-Jun-2006 to 29-Jul-2006. Each of the nine sample plots was 30 × 220 m. Multiple collecting methods were used to ensure broader taxonomic representation because different methods are biased toward different behavioural categories of insects. Sweeping vegetation primarily captures specimens on vegetation (e.g. phytophages, predators). Yellow pan traps collect flying individuals attracted by bright colours (e.g., pollinators and other flower visitors) as well as flies walking on the ground surface (e.g., saprophages on the substrate, weak flyers). Malaise traps intercept flying insects that move upward upon encountering an obstacle, so strong flyers are more likely to be captured. Two sweep transects were sampled every 7–8 days using 60 sweeps of a sweep net. Five yellow pan traps (plastic bowls placed in the soil with their upper rim flush with the ground surface and filled with salt water, propylene glycol and a drop of liquid detergent as a wetting agent) were placed 20 m apart on a 80 m transect in the center of each site and emptied every 7–8 days. One Malaise trap was also installed in each site for three consecutive days in weeks 2, 5 and 8.

Insects were preserved in 70% ethanol prior to chemical drying and mounting for identification. Specimens are deposited in the Lyman Entomological Museum (McGill University, Ste-Anne-de-Bellevue, QC). All specimens of

Brachycera, except the taxonomically difficult families Phoridae and Anthomyiidae, were identified to named species if keys or taxonomic expertise were available, or to numbered morphospecies.

Functional diversity (Petchey and Gaston 2006) was assessed in two ways. Species were divided into three size classes: small (<5 mm), medium (5–10 mm) and large (>10 mm) and were also placed in one of six trophic groups and 14 subgroups (Online Resource 1; Beaulieu and Wheeler 2001) based on available taxonomic and ecological literature (e.g., Ferrar 1987; Blades and Marshall 1994). The trophic group was based on larval habits because that is the dominant feeding stage. Species not assigned to a trophic subgroup, either because they were reared from a variety of media or because their precise habits were unknown, were pooled in their main trophic group. Species for which the trophic group was unknown were excluded from analyses.

## Habitat and vegetation variables

Vegetation cover was quantified using the Braun-Blanquet scale: 0 (<1%); 1 (1–5%); 2 (6–25%); 3 (26–50%); 4 (51–75%); 5 (76–100%; Goldsmith et al. 1986) for each of nine strata: *Sphagnum* mosses, other mosses, lichens, herbs, ericaceous shrubs, grasses and sedges, horsetails, bare peat and open water. Soil moisture was categorized as follows: 0 (very dry, soil cracking); 1 (dry); 2 (moist); 3 (water table at surface); 4 (water table above surface).

One peat sample (25 cm × 25 cm × 3 cm deep) was collected in the center of each site. A 250 ml subsample was separated for pH analysis and saturated with deionised water at room temperature. Fresh homogenized material for soil analysis was immediately frozen. For N (as $NO_3$ and $NH_4$), a KCl extraction was performed on wet samples and analysed by colorimetry. For P, K, Ca, Mg, Fe, Mn and Na, defrosted samples were dried at 65°C in an air oven for 2–3 days.

## Statistical analysis

Because pan traps were occasionally disturbed by animals or wind, analyses were based on four pan trap replicates per treatment per week. For weeks in which five pan trap samples were available, one trap per treatment was randomly omitted from analyses. Species abundance data from each trap type (Malaise trap, pan trap, sweeping) in each sampling week were pooled for each treatment per site. Intra-site comparison was done to compare diversity, dominance and composition in the three treatments, to account for regional variation in species composition.

Individual-based rarefaction curves (Gotelli and Colwell 2001) based on 1,000 permutations with species richness as a diversity index were generated using ECOSIM version 7.0 (Gotelli and Entsminger 2001). Species richness overall and of the dominant trophic groups of each treatment per site were calculated using rarefaction estimates standardised to the lowest number of individuals collected in any of the treatments in each site. Total estimated species richness was calculated in each treatment per site using the non-parametric abundance based coverage estimator (ACE), using EstimateS version 7.5 (Colwell 2005). Simpson's diversity index was also calculated as a dominance measure using EstimateS.

Community composition (differentiation or similarity of species assemblages in sites) was analysed based on log-transformed (log (x + 1)) relative abundance of species represented by three or more individuals. All Brachycera were analysed, as well as three subordinate taxonomic groups: Lower Brachycera + Aschiza; Acalyptratae; and Calyptratae. Subordinate taxa were analysed separately because phylogenetically related taxa often have similar life-history and resource bases (Tokeshi 1993). Community composition was also analysed for each size class and trophic group. Community composition among the three treatments was compared using non-metric multidimensional scaling (NMDS), or cluster analysis if no NMDS solution was found, and multi response permutation procedures (MRPP). For NMDS, an initial 6-dimensional analysis was done, stepping down in dimensionality until the number of ordination axes was sufficient to achieve low stress values. For the final ordination, the *n*-dimensional Sorenson ordination with 500 iterations was used as the starting configuration. Bray-Curtis distance and the group average distance method were used for cluster analysis. A Sorenson distance metric was applied to each MRPP, in a similar fashion to NMDS or cluster analysis to test for pair-wise differences in species composition between treatment types. Analyses were performed using PC-ORD version 4.36 (McCune and Mefford 2005).

Indicator species analysis (Dufrêne and Legendre 1997) was performed on log-transformed abundance of species with three or more individuals using PC-ORD to determine associations of particular species or trophic groups with treatment type. Species or trophic groups with more than 15 individuals, a significant p-value ($\leq 0.05$) assessed using a Monte Carlo randomization test based on 1,000 permutations and an indicator value (IndVal) greater than 45 were considered as indicators for a treatment.

Environmental variables were analysed with NMDS and MRPP to assess patterns of site relationships for vegetation and peat chemical composition. To reduce the number of environmental variables, two principal component analyses (PCA) were performed to condense the peat chemistry and vegetation cover variables. Cross-products matrices containing Pearson correlation coefficients were used to produce a standardized PCA. From each PCA, the three axis scores explaining most of the variance were used as independent variables in the subsequent multivariate analysis. Log-transformed relative abundance of species represented by three or more individuals and log-transformed abundance of the 18 trophic subgroups were analysed using Canonical Correspondence Analysis (CCA) in relation to eight variables (Peat1, Peat2, Peat3, Vege1, Vege2, Vege3, pH, field moisture). Axis scores were centered and standardized to unit variance. A Monte Carlo test based on 200 runs was used to assess the significance of the axis eigenvalues. The null hypothesis was that there is no relationship between the environment and species matrices. Multivariate analyses were performed using PC-ORD.

## Results

### Taxonomic diversity

A total of 20,653 specimens representing 699 named species and morphospecies were used in analyses (Online Resource 1). Between 39 and 50% of the species were represented by only one specimen in the treatments. The rarefaction curves for all treatments per site did not reach an asymptote (Fig. 1), and ACE suggested that 59–85% of the species present were collected.

More specimens were collected in abandoned treatments, followed by restored and natural treatments (Table 1). Simpson's index was lower in each of the abandoned treatments indicating that one or two species constituted a large proportion of the total abundance. In Bois-des-Bel, the diversity (rarefaction estimate) of the restored treatment was not significantly different than that in the natural treatment, but was significantly higher than that in the abandoned treatment. In Chemin-du-Lac, diversity in the restored treatment was significantly lower than that in the natural and abandoned treatments, which did not differ significantly from one another. In St-Charles, diversity was highest in the restored treatment, followed by the natural and abandoned treatments (Table 1). In indicator species analysis, *Paroxyna albiceps* (Loew) (Tephritidae) (IndVal $= 90$, $P = 0.035$) and *Minettia lupulina* Fabr. (Lauxaniidae) (IndVal $= 48$, $P = 0.035$) had significant associations with natural sites. *Oscinella* sp. A (Chloropidae) (IndVal $= 63$, $P = 0.011$), *Discocerina obscurella* (Fallén) (Ephydridae) (IndVal $= 100$, $P = 0.037$) and *Scatella stagnalis* (Fallén) (Ephydridae) (IndVal $= 76$, $P = 0.037$) were associated with abandoned sites; no species were associated with restored sites.

The species assemblages of the three treatments in each site were clearly grouped in the cluster analysis (Fig. 2) and this was supported by MRPP (Table 2). However, the overall assemblage of species was significantly different between the three peatlands ($P = 0.0013$).

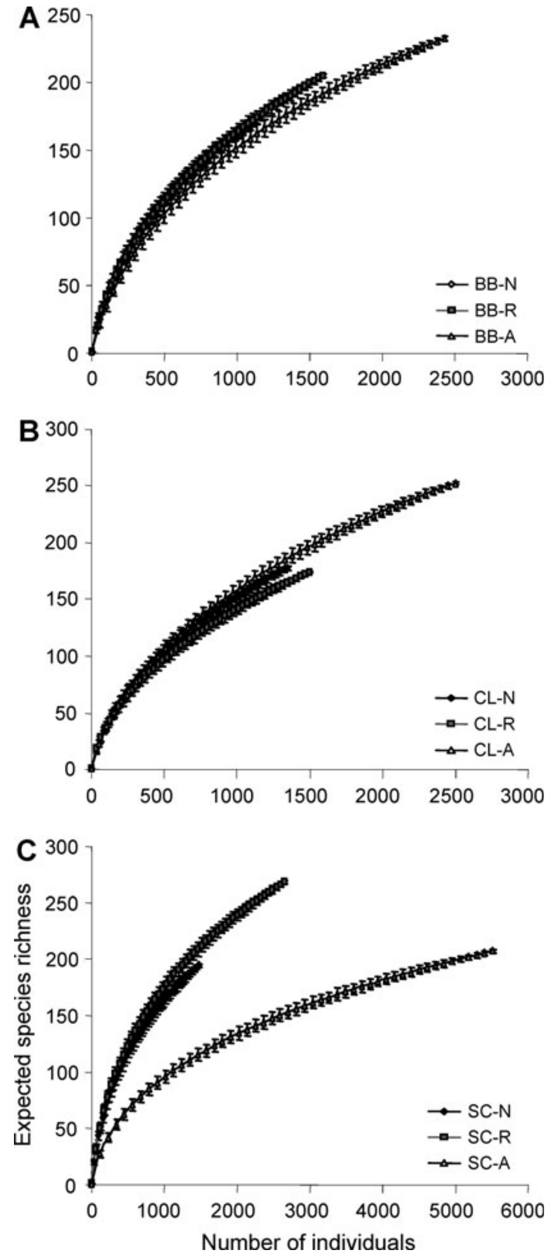


**Fig. 1** Rarefaction estimates of expected species richness (± 1 SD) of Brachycera plotted against number of individuals for natural (*N*), restored (*R*) and abandoned (*A*) treatments in **a** Bois-des-Bel bog, **b** Chemin-du-Lac bog and **c** St-Charles bog

In separate analyses of subordinate taxonomic groups, acalyptrate flies showed a different pattern of distribution among treatments; assemblages were similar in restored and abandoned treatments, but significantly different from both in natural sites, based on MRPP. In the acalyptrates, 93% of the species were in the small size class. Calyptrate assemblages were significantly different only in natural and abandoned treatments. Lower Brachycera + Aschiza assemblages were similar among all treatments (Table 2).

**Table 1** Raw species richness ($S_{obs}$), number of individuals (N), rarefaction estimated species richness ($S_{est}$) (species ± SD, standardized at 1,100 (Bois-des-Bel), 1,300 (Chemin-du-Lac) and 1,400 (St-Charles) individuals) and Simpson's diversity indices of Brachycera in natural (N), restored (R) and abandoned (A) treatments in the three study sites: Bois-des-Bel (BB); Chemin-du-Lac (CL) and St-Charles (SC)

| Site | $S_{obs}$ | N | $S_{est}$ | Simpson |
|---|---|---|---|---|
| BB-N | 177 | 1,198 | 169.94 ± 2.42 | 19.78 |
| BB-R | 207 | 1,629 | 171.31 ± 4.87 | 17.38 |
| BB-A | 235 | 2,486 | 159.34 ± 5.73 | 8.39 |
| CL-N | 180 | 1,389 | 174.20 ± 2.3 | 12.14 |
| CL-R | 176 | 1,531 | 162,15 ± 3.3 | 15.62 |
| CL-A | 255 | 2,559 | 182.82 ± 6.18 | 9.03 |
| SC-N | 197 | 1,514 | 190.57 ± 2.3 | 15.94 |
| SC-R | 271 | 2,719 | 205.79 ± 5.76 | 16.08 |
| SC-A | 209 | 5,628 | 112.85 ± 5.88 | 3.53 |

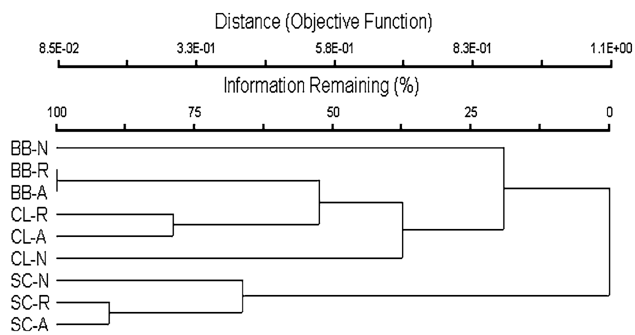


**Fig. 2** Cluster dendrogram based on log-transformed relative abundance of Brachycera assemblages in natural (*N*), restored (*R*) and abandoned (*A*) treatments in the study sites

**Table 2** *P*-values for MRPP pairwise comparisons among natural (N), restored (R) and abandoned (A) treatments for Brachycera separated by subordinate taxa, size class and trophic habits

| Treatments | *P* (N-R) | *P* (R-A) | *P* (N-A) |
|---|---|---|---|
| All Brachycera | 0.32 | 0.69 | 0.080 |
| Acalyptratae | **0.027** | 0.28 | **0.025** |
| Calyptratae | 0.28 | 0.78 | **0.043** |
| Lower Brachycera + Aschiza | 0.75 | 0.82 | 0.54 |
| Small size | 0.069 | 0.52 | **0.035** |
| Medium size | 0.69 | 0.76 | 0.40 |
| Large size | 0.84 | 0.87 | 0.86 |
| Trophic habits | **0.029** | 0.42 | **0.036** |

Significant differences ($P \leq 0.05$) in species composition between treatments are in bold

## Functional diversity

Within each treatment, most individuals were in the small size class, followed by medium and large classes (Fig. 3). The relative abundance of the small size class decreased from abandoned to restored to natural treatments. There were significant differences in species composition in small species among natural and abandoned treatments and differences were marginally significant between natural and restored treatments. The composition of the small size class was similar in restored and abandoned treatments (Table 2). Species richness of the small size class in restored treatments was either intermediate to that in natural and abandoned treatments or higher than in natural treatments. The higher number of specimens in abandoned treatments was due to *Scatella stagnalis* and *Chrysotus* spp. (Dolichopodidae) which made up a large proportion of the Brachycera assemblage in those sites (Online Resource 1).

Trophic assemblages were dominated by predators and saprophages, followed by phytophages (Online Resource 1); there were relatively few parasites or omnivores. The relative abundance of saprophages and predators decreased from abandoned, to restored to natural treatments (Fig. 4). Species diversity of predators in restored treatments was either not significantly different from the other treatments (Bois-des-Bel), intermediate between abandoned (highest diversity) and natural (lowest diversity) sites (Chemin-du-Lac) or significantly higher than natural and abandoned sites which were not significantly different (St-Charles). Although saprophages were more abundant in abandoned treatments, species richness was lower than in restored and natural treatments, particularly in St-Charles (Table 3). The lower evenness in abandoned treatments was due to two dominant saprophagous species, *Scatella stagnalis* (Chemin-du-Lac, St-Charles) and *Paramyia nitens* (Loew) (Milichiidae) (Bois-des-Bel). Only substrate saprophages (SAsu) showed discrimination among treatments; with a significant association with abandoned treatments (IndVal = 54, $P = 0.029$). Trophic assemblages in restored and abandoned treatments were not significantly different, but both differed significantly from natural treatments (Table 2).

## Community composition and ecosystem properties

The three treatment types differed significantly in environmental variables (Online Resource 2, 3) based on NMDS (Fig. 5) and MRPP ($P = 0.001$), with conditions in restored treatments intermediate to those in natural and abandoned treatments.

The first three axes of the PCA on peat chemical data explained 80% of the variance. The first three axes of the PCA on vegetation cover data explained 82% of the variance. Thus, little of the information on environmental variables was lost with the three axes extracted from the PCA reduction. Those six PCA-axis scores along with peat pH and field moisture were used as independent variables

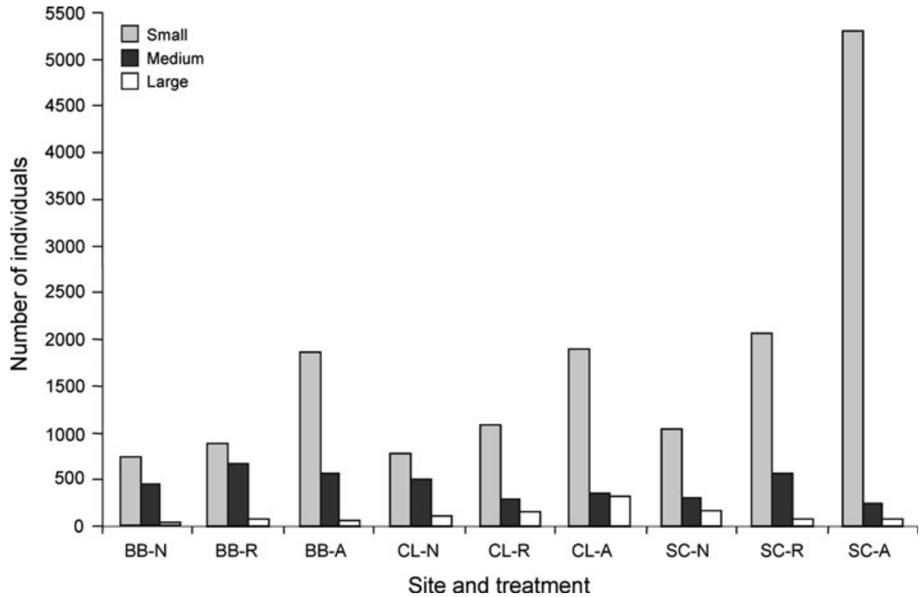


**Fig. 3** Relative abundance by size classes of Brachycera collected in each treatment in the study sites

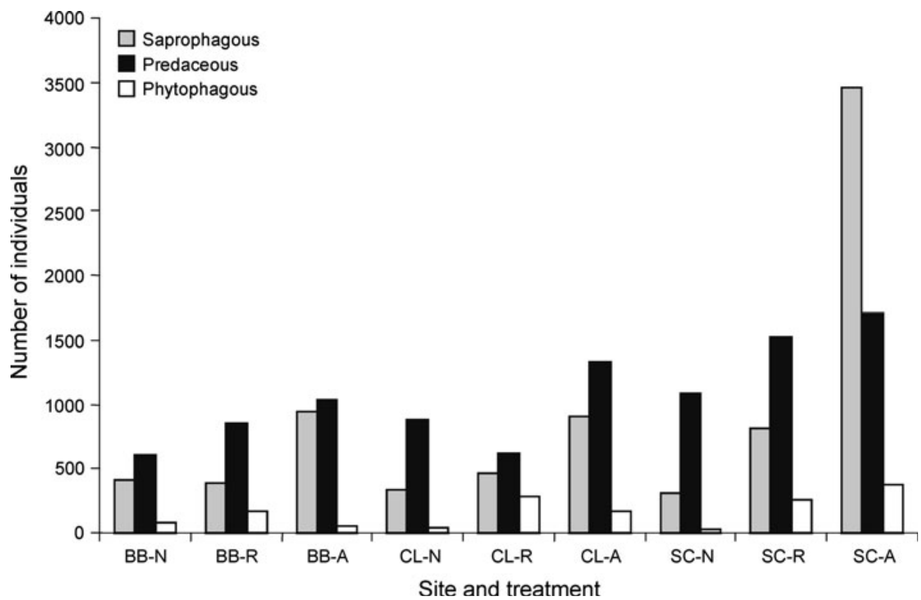


**Fig. 4** Relative abundance of the three dominant trophic groups (saprophagous, predaceous, phytophagous) for Brachycera in each treatment in the study sites

in the CCA to determine the combination of variables that best explain the species and trophic distribution among the treatments.

The CCA explained 55.4% (axis 1: 23.7%; axis 2: 18.1%; axis 3: 13.6%) of the variance in Brachycera species composition (Fig. 6). Axis 1 mostly separated sites by Peat3 and Vege3, axis 2 by Peat1 and Vege1, and axis 3 by pH. Most of the assemblages in restored and abandoned treatments were negatively related to axis 2, indicating a preference for a high coverage of bare peat where P, K and Zn concentrations were lower. Assemblages in natural treatments were associated with a high coverage of *Sphagnum* and ericaceous shrubs.

The CCA explained 79.2% (axis 1: 35%; axis 2: 28.8%; axis 3: 15.4%) of the variation in trophic composition (Fig. 7). Data are plotted on axes 1 and 3 to show the primary environmental gradient driving the separation of trophic assemblages. Axis 1 was positively related with Peat3 and Vege1; axis 2 was negatively related with Vege1 and positively with Vege3; axis 3 was positively related with Peat1 and Vege3 and negatively with pH. Predaceous flies in dung (PRco) were mostly affected by peat chemical properties (Peat3). Mollusc predators (PRmo) were mostly affected by the coverage of bare peat (Vege1) and peat pH. Leaf litter saprophages (SAlf) were positively associated with a high coverage of herbs and trees where mineral nutrients are at higher levels. Necrophagous saprophages (SAne) were negatively associated with herbs and trees (Vege3) and positively with coverage of *Sphagnum* and

**Table 3** Rarefaction estimates of species richness (species ± SD) for predaceous (standardised at 330 (Bois-des-Bel), 480 (Chemin-du-Lac) and 685 (St-Charles) individuals) and saprophagous (standardised at 350 (Bois-des-Bel), 300 (Chemin-du-Lac) and 300 (St-Charles) individuals) trophic groups in natural (N), restored (R) and abandoned (A) treatments in the three study sites: Bois-des-Bel (BB); Chemin-du-Lac (CL); St-Charles (SC)

| Site | Predator | Saprophage |
|---|---|---|
| BB-A | 58.04 ± 4.05 | 43.54 ± 3.23 |
| BB-R | 52.91 ± 3.64 | 55.77 ± 1.45 |
| BB-N | 55.62 ± 3.09 | 49.14 ± 1.73 |
| CL-A | 67.99 ± 4.19 | 37.22 ± 3.29 |
| CL-R | 61.0 ± 2.53 | 47.26 ± 2.77 |
| CL-N | 53.8 ± 3.29 | 45.76 ± 1.35 |
| SC-A | 73.66 ± 3.65 | 17.86 ± 2.59 |
| SC-R | 93.57 ± 3.69 | 49.38 ± 3.38 |
| SC-N | 77.91 ± 3.08 | 48.32 ± 0.8 |

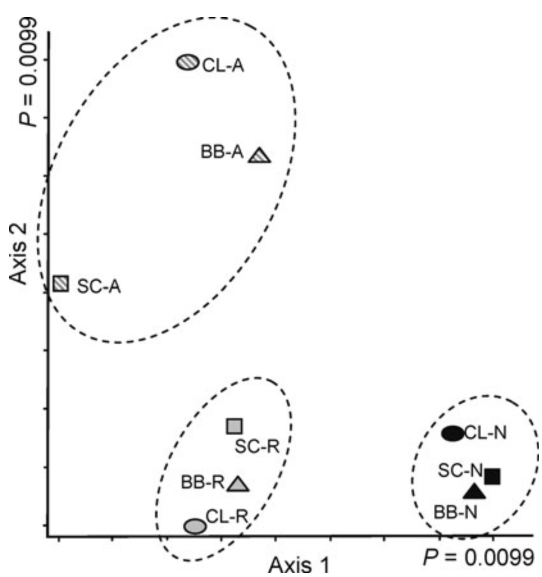


**Fig. 5** Non-metric multidimensional scaling ordination of environmental variables based on vegetation cover, peat chemistry, pH and soil moisture in the three treatments. The axes of a two-dimensional solution are plotted. *Symbols* represent variables in natural (*N*), restored (*R*) and abandoned (*A*) treatments in the study sites. Ordination differs from randomly derived matrices at $P < 0.05$ (Monte-Carlo test, 100 permutations)

ericaceous shrubs (Vege1). Substrate saprophages (SAsu) were associated with higher coverage of bare peat. Fungivores (SAfu) were mostly affected by peat chemical properties (Peat3) and the type of vegetation (Vege1). Patterns in stem-borers (PHsb) and flower consumers (PHfl) were mostly predicted by moderately high coverage of bare peat and moderately low cover of herbs and trees. The other groups were close to the centre of the biplot and thus more evenly affected by a set of environmental variables.

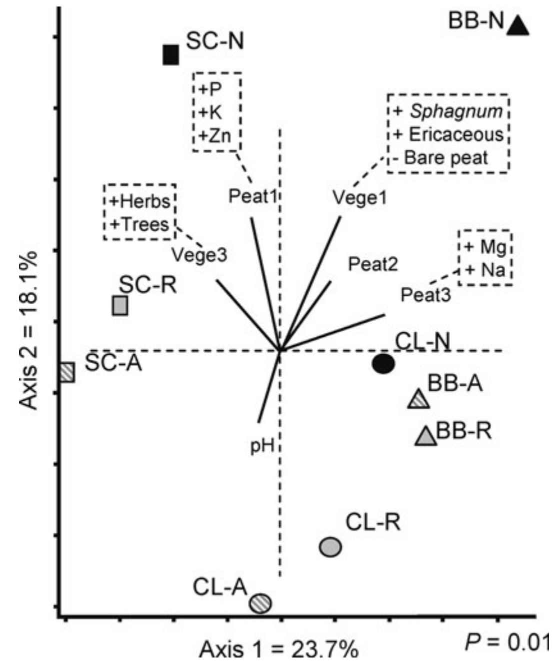


**Fig. 6** Canonical correspondence analysis of Brachycera (axes 1 and 2 shown, individual species omitted for clarity). Environmental vectors are shown as *solid lines*. Variables with highest eigenvector values for PCA reduction (positive and/or negative correlations) are in *dashed boxes* beside vectors. Percent variance and stress values are shown on axes

## Discussion

### Effects of peat mining and restoration

The treatments contained species with a range of environmental requirements and tolerances to changing conditions. This ecological diversity illustrates the advantage of species-level identification of insects: many higher taxa (orders, families) contain species with a broad range of tolerance and ecological roles. Habitat changes favour some species and disfavour others, causing compensatory shifts in species abundance (Brown et al. 2001). Species evenness was lower in abandoned treatments than in natural and restored treatments, and species richness was also lower in abandoned sites at Bois-des-Bel and St-Charles. The lower species richness in the abandoned site at St-Charles may be due to the dominance of *Scatella stagnalis*, which thrives on bare peat (Foote 1995). The presence of that species probably affected the occurrence and colonization patterns of other species, especially small saprophagous acalyptrates that might compete with *S. stagnalis* for detritus and algae. Low species richness of saprophages in abandoned areas was compensated by high abundance of a few species. This was supported by indicator species analysis, in which *S. stagnalis* was strongly associated with abandoned areas and made up the highest

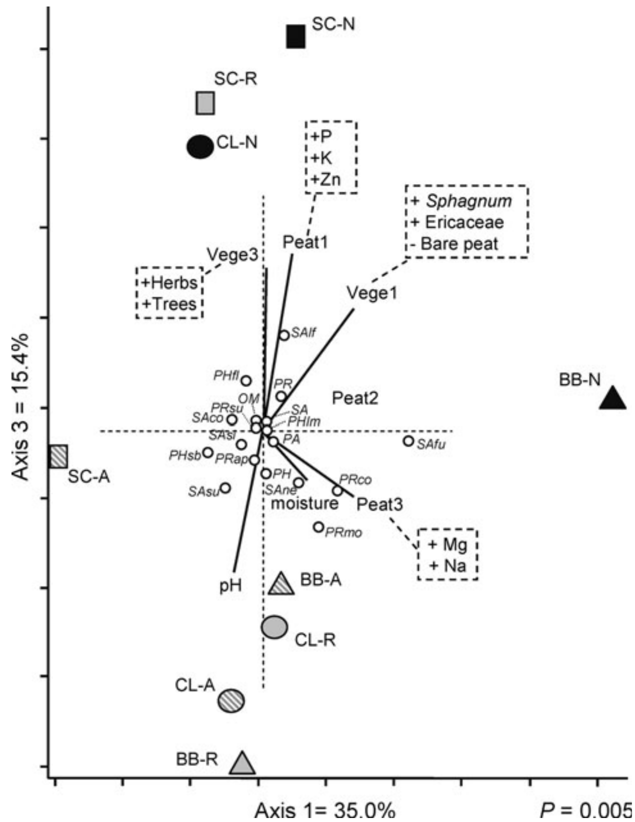


**Fig. 7** Canonical correspondence analysis of trophic assemblages (axes 1 and 3 shown). Environmental vectors are shown as solid lines. Variables with highest eigenvector values for PCA reduction (positive and/or negative correlations) are in *dashed boxes* beside vectors. Percent variance and stress values are shown on axes

proportion of individuals in the SAsu indicator group. New conditions caused by peat mining are probably a subset of the pre-existing conditions in natural sites, and species favoured by disturbance were likely present in natural sites and may have colonized abandoned sites quickly, becoming established earlier than other species.

Restoration of native characteristics in previously mined areas resulted in the re-establishment of diverse Brachycera assemblages and normal community structure. The species abundance distributions were similar in natural and restored sites and restored sites were distinct from abandoned sites. Perturbation can reset the successional stage of a community and similar patterns should emerge in increasingly stable environments (Death 1996). Higher diversity was observed with increased time after restoration, as found in other studies (Jansen 1997; Waltz and Covington 2004; Summerville et al. 2007).

In contrast, guild structure was altered in restored areas, in which trophic composition was most similar to abandoned areas. This was most pronounced in saprophages and predators. The species in those groups were very abundant in abandoned treatments and this was also seen at an intermediate level in restored treatments. However, there was successful recolonization of saprophagous and predaceous species in restored areas, as suggested by the species richness estimates. Dunn (2004) found that recovery of species composition was slower than that of species richness. Williams (1993) found similar results in restored riparian woodlands; some functional groups responded as quickly as 3 years, while others did not. As the restored sites in our study age and progress toward a natural bog, the Diptera fauna will likely converge on that of nearby natural sites. Watts et al. (2008) found that age and vegetation structure complexity were the major factors influencing convergence of beetle diversity in restored and undisturbed peatlands. A feedback exists between species/trophic composition and ecosystem processes, and these processes will recover on different time scales after restoration (Palmer et al. 1997). It is estimated that characteristic bog plant species can be established in 3–5 years following restoration, normal hydrology in 10 years and the peat accumulation system in 30 years (Rochefort et al. 2003). Our results suggest that 7 years were sufficient to recover a Brachycera species community and a level of diversity (in two of the bogs) characteristic of a natural bog, but not trophic composition or the small sized species assemblage. It is difficult to determine the time needed for complete recovery of invertebrate species assemblages in restored wetlands; estimates range from 4 to 17 years (Streever et al. 1996; Stanczak and Keiper 2004).

It is often assumed that habitat restoration will lead to natural re-colonization of small invertebrates, fungi and microbes without further facilitation (e.g., Moynahan et al. 2002; Andersen et al. 2006; Mazerolle et al. 2006). In our study, disturbance had the greatest effect on the distribution of small Brachycera and acalyptrate flies (which made up the great majority of the small size class); many of these species may be dispersal-limited because several acalyptrates do not appear to fly readily or for long distances (T.A. Wheeler, *personal observations*). Because those species experience the landscape on a small spatial scale, they may be more susceptible to changes in resource availability. Kumssa et al. (2004) found that although rehabilitation could enhance post-mining recovery of soil micro-arthropods, it would take a long time. In that study, as in ours, colonization potential is a limiting factor for recovery of assemblages.

Some Diptera species are abundant and characteristic in peatlands, probably because their larval or adult food resource or breeding media are restricted to or characteristic of peatlands. Marshall (1994) identified 73 species of Sphaeroceridae in Canadian peatlands of which 15 were considered characteristic of peatlands. Three of those species were found in this study: *Spelobia pappi* Roháček and *Phthitia ovicercus* Marshall in abandoned and restored sites; and *Dahlimosina dahli* (Duda) in natural sites. Smaller numbers of species in other families are also known to be associated with peatlands. Two species of Canadian Sarcophagidae, *Fletcheromyia fletcheri* (Aldrich) and *Sarcophaga sarraceniae* Riley, are indicators of the peatland-restricted pitcher plant *Sarracenia purpurea* (Farkas and Brust 1986) and were found in the three treatments. In the Tabanidae (Teskey 1969, Teskey and Burger 1976), a number of species of *Hybomitra* Enderlein considered characteristic of bogs were found in our sites: *H. minuscula* (Hine), *H. sodalis* (Williston), *H. trepida* (McDunnough), *H. typhus* (Whitney) and *H. pechumani* Teskey and Thomas. Several species of *Tabanus* L. are associated with peatlands, but only *T. novaescotiae* Macquart was found in restored and abandoned sites. In the Lauxaniidae, Miller (1977a, b) considered *Homoneura sheldoni* (Coquillett) a usual inhabitant of bogs. In the Empididae, many species of the genus *Rhamphomyia* Meigen appear to be characteristic of peatlands (Barták and Roháček 1999), although many of the Canadian species are undescribed and existing keys are inadequate. In addition to these species, there is probably a large number of peatland associated Diptera still to be documented and the status of many species is unknown because of insufficient ecological knowledge.

### Effects of environmental conditions

Vegetation composition and substrate quality influenced patterns in species and trophic groups among the treatments at the site scale, thus the distinct assemblage in restored sites compared to natural sites may be a consequence of the intermediate environmental conditions.

Because the restored areas had not yet approached natural peatland conditions, environmental conditions amenable to saprophages and predators, especially high pH and the presence of bare peat, were still dominant. The distribution of saprophages was mostly affected by substrate quality. The muddy surface in abandoned areas likely accounted for higher success of deposit feeding and algivorous Ephydridae such as *Scatella stagnalis* and *Discocerina obscurella* in contrast to the compact and less accessible substrate in natural areas. The composition of peatland litter changes radically after drainage; from *Sphagnum* moss, sedges and shrubs to essentially bare peat (Laiho et al. 2003). Increased aeration within the upper peat layer exposes organic matter to aerobic microbial activities (Vasander and Laiho 1995; Silins and Rothwell 1999), increasing the amount of material available for saprophages. Silvan et al. (2000) and Laiho et al. (2001) found changes in abundance and community composition of soil animals following drainage, with the number of invertebrates positively correlated with a lower water table. In contrast, predators were present where their prey was most likely to be found.

Both indicator species in natural sites are associated with particular plants. *Paroxyna albiceps* feeds on multiple species of *Aster* (Asteraceae; Novak and Foote 1968), which were absent or rare in restored sites and apparently have not recolonized. *Minettia lupulina* mines dead leaves of multiple species of trees (Miller 1977a, b), including black spruce (*Picea mariana*) a dominant bog species in the region (Pellerin and Lavoie 1999).

## Conservation implications and conclusions

The similarity in species composition among the treatments could be attributed to resilience (ability to reorganize after disturbance) of Brachycera in a mosaic of well connected habitat types. The natural areas probably act as a source of colonists, because the three site types were within one km of each other. Species with high dispersal ability were able to recover from constructive (restored sites) and destructive (abandoned sites) human activities; however, dispersal-limited species did not recover as quickly. Accordingly, maintaining natural bog habitats near restored sites is desirable to facilitate colonization and regeneration. Those natural areas would be best directly connected to restored sites without physical barriers to dispersal so as to maximize connectivity between populations (Scott et al. 2001; Tscharntke et al. 2005). In Bois-des-Bel and Chemin-du-Lac, restored and natural sites were adjacent, but separated by an edge of dense trees and drainage ditches. When restored sites are isolated from a colonist source, corridors or facilitation may encourage re-establishment. Gilbert et al. (1998) found that corridors between fragmented microecosystems increased arthropod immigration rates and species richness compared to disconnected habitat patches. Brown et al. (1997) and Brady et al. (2002) investigated whether inoculation (indirect addition of invertebrates in a soil sample from a natural site, equivalent to the restoration technique used in this study) and/or stocking (direct addition of specimens) of dispersal limited invertebrates could facilitate establishment in restored sites. In the short term, facilitation increased recruitment of some taxa, mainly Gastropoda, Odonata, Ephemeroptera, Hemiptera, Coleoptera and Diptera.

Ecosystems are spatially and temporally heterogeneous and organisms respond differently to restoration; thus, although restoration success is often claimed, defining success is not always straightforward. In addition, the nature of restoration means that many studies have only one site per treatment (e.g., Williams 1993; Armitage et al. 2006), and the results are applicable only to that site, limiting the ability to extrapolate about mechanisms of restoration success on broader scales (West et al. 2000).

The replication in this study, with multiple restored bogs, means that observed patterns are less likely due to idiosyncracies of a particular site. The proximity of abandoned mined and natural sites to restored treatments also allowed the use of positive and negative controls. Comparison of restored sites should be based, where possible, on more than one reference site (Ruiz-Jaen and Aide 2005) ranging from those with high levels of function (e.g., natural bogs) to those that are highly disturbed (e.g., abandoned mined sites) (Brinson and Rheinhardt 1996). This has the advantage of documenting community resilience as well as the status of restored sites when full recovery is not achieved.

The desired endpoint of restoration must be determined when assessing success: recovery of a species community similar to that before disturbance; or recovery of a trophic structure regardless of the constituent species. Despite studies on a variety of scales and taxa, we still lack knowledge of which species or trophic groups really matter to natural peatland function, and whether the presence of particular species accelerates or impedes recovery. Biotic interactions may induce variability between diversity and ecosystem function (Peterson et al. 1998) and the sequence of species following restoration may have a major influence on ecosystem performance (Elmqvist et al. 2003). Passive restoration projects need information on the distribution, dispersal and demography of species (Scott et al. 2001). Many insects are bog-restricted or bog-associated, although the lack of taxonomic resolution, and ecological knowledge does not allow all those species to be distinguished (Spitzer and Danks 2006). Given the ongoing decline of peatlands, such knowledge is central to conservation and restoration. Although restoration is an important tool for regenerating peatland biodiversity, conservation of pristine lowland peatlands is more valuable for maintaining peatland biota, biodiversity and ecosystem function.

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